Forgetting User Preference in Recommendation Systems with Label-Flipping - Implementation Project

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I. Introduction

The problem our project will address is that most recommendation systems cannot forget user data efficiently. With this, it loses accuracy or uses excessive computing resources.To resolve this, the paper mentions a solution that works by flipping user data and fine tuning the model to make it faster, efficient and effective. Our project aims to implement a recommendation system capable of efficiently forgetting user data in compliance with privacy requirements, while maintaining high recommendation accuracy and minimizing computational expenses. The primary objective is to reproduce and integrate the algorithm proposed in the paper "Forgetting User Preference in Recommendation Systems with Label-Flipping" using two core models: Neural Collaborative Filtering (NCF) and Deep Matrix Factorization (DMF). Through this implementation, we seek to demonstrate how targeted user data can be effectively removed without the need for full model retraining, ensuring both system efficiency and performance preservation.

II. Technical

1) Algorithms

The algorithms that was used in this paper was unlearning model:

Retrained (Baseline)- Deletes the targeted user's data and re-trains the entire model from scratch using remaining data.

SISA [1]- Splits data into multiple random partitions (shards), trains a separate submodel on each shard, makes predictions by averaging results from all submodels, and if need to forget a user, only retain submodels where that user appeared.

RecEraser [2]- Improvement on SISA instead of random shards it is grouped by similarity and preserves collaborative patterns better.

BadT [3]- Use student-teacher model- taking the original recommendation model as teacher and a randomly initialized model as student. Students selectively take knowledge from the teacher to reach the target forgetting specific infomation.

erMax [4]- Modifies user's data into noisy versions, and fine-tunes maximize errors on these users, and repair steps then fine-tunes the model on valid users to restore accuracy.

Training Model:

NCF [9]- This is a deep learning-based recommendation model that predicts how likely a user is to interact with a specific item. It begins by converting user IDs and item IDs into embedding vectors that capture the latent features of both users and items. These vectors are then concatenated and passed through a multi-layer perceptron (MLP), which learns complex non-linear relationships between users and items. The final output is a score between 0 and 1, representing the predicted preference of the user for the item.

DMF [8]- This is based on a nonlinear latent variable model which essentially uses a statistical model that uses unobserved variables to explain observed data making it nonlinear rather than linear as the relationships between the latent and observed are what is being compared. Matrix factorization overall is a simple embedding model and an advanced method for breaking down complex data into multiple layers or simple components. DMF extends this concept by introducing multiple layers of factors so it allows extraction based on hierarchy within the data. This is useful for uncovering very intricate or interrelated structures that single-layer factorizations might tend to miss.

FlipRec [5]:

Targeted Users Label-Flipping- The goal of this step is to create anti-samples that help the model forget what it previously learned about specific targeted users. This is achieved by flipping their interaction labels—likes are turned into dislikes and vice versa. These flipped samples, known as Dflipf, act as anti-samples designed to remove the influence of the targeted users from the model.

Contextual User Augmentation- To ensure that the model's performance remains stable for non-targeted users after label flipping, this step introduces additional data. It identifies a group of users (Us) who have interacted with the same items as the targeted users. For each of these users, their positive interactions with those shared items are collected, and k negative samples are added to balance the data. This new dataset, called Dtrs, helps preserve valuable information for the rest of the user base.

Fine-Tuning the Model- The final step involves updating the model using both Dflipf and Dtrs. The model is fine-tuned using binary cross-entropy loss to learn from the updated labels. Additionally, knowledge distillation (KD) loss is applied to ensure the new model's predictions remain similar to the original model for non-targeted users. This is done by minimizing the mean squared error (MSE) between the outputs of the two models. The total loss function used for training, called LFlipRec, is a combination of both: LFlipRec = Lrec + LKD

2) Analysis

Among the unlearning methods evaluated, FlipRec stands out as the most effective in balancing the trade-off between forgetting and retention. Retrained serves as a baseline to remove the targeted user's data and retraining the model from scratch is very expensive to do. SISA improves efficiency by partitioning the dataset into shards and training the submodels, but it uses random splits which can weaken collaborative filtering. RecEraser enhances SISA by grouping users based on similarity, which better preserves collaborative patterns and improves retention. BadT uses a student-teacher framework to selectively forget user-specific information while preserving overall model performance. erMax modifies user data with noise and maximizes errors on those users before repairing the model with valid data. Each of the unlearning methods have strengths but FlipRec archives overall balance of forgetting targeted users while retaining knowledge for non-targeted users with minimal accuracy loss.

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DMF and NCF are both deep learning models used in recommendation systems. NCF takes user IDs and item IDs, turns them into embeddings (features) and passes them through a neural network to learn how users and items relate. It is good at learning complex patterns and works well when there's not a clear relationship between users and items. DMF builds on matrix factorization but adds more layers to make it deeper. It tries to understand hidden factors behind user behavior by breaking data down into multiple layers. Doing this helps DMF find deeper and more structured patterns. In short, NCF is better for learning flexible and detailed user-item interactions and DMF is better at uncovering deeper, layered relationships in the data.

3) Implementation

All datasets are of MovieLen provided in the paper and converted to implicit feedback, turning detailed rating into binary labels of 0 or 1 to indicate user-item interaction. The users were splitted into two groups: 5% as targeted users and 95% as retained users. Switch into three groups of train, test, and validation and run on a sample size of 10,000 on google colab A100 CPU or T4 CPU. Each is run five times and averaged out to get final results.

Hyperparameters Settings: We used Adam optimizer for training, learning rate is 0.00015 and batch size 64. DMF was conducted over a single epoch but the NCF model used 2 epochs. We used one negative sample of k = 1. NCF we used hidden layers with dimensions of 32-16-8-1. DMF dimensions of 512-64 for users and 1024-64 for items.

III. Evaluation

For both groups, evaluation metrics focuses on whether the unlearning baselines can achieve recommendation performance comparable to that of a fully retrained model. We used Area Under ROC Curve (AUC), Hit Ratio (HR), and Normalized Discounted Cumulative Gain (NDCG).Additionally, to evaluate the similarity in recommendation output between unlearning and retrained models we used Activation distance and JS-Divergence.

NCF Results

Retained Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.509818	0.900685	0.986395	1.0	0.525928	0.538549	0.541616
1	SISA	0.502946	0.926654	0.975809	1.0	0.594039	0.619309	0.695642
2	RecEraser	0.528836	0.902465	0.982621	1.0	0.537761	0.542237	0.537691
3	BadT	0.540428	0.895548	0.982993	1.0	0.551373	0.555490	0.572588
4	erMax	0.551924	0.907534	0.986395	1.0	0.561946	0.563338	0.573397
5	FlipRec	0.503088	0.893836	0.972789	1.0	0.528105	0.532354	0.544608

Targeted Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.526864	0.827586	1.0	1.0	0.518400	0.572982	0.696620
1	SISA	0.565009	0.933333	0.2	0.0	0.776809	0.173409	0.000000
2	RecEraser	0.584558	0.775000	1.0	0.4	0.551480	0.559602	0.283043
3	BadT	0.588655	0.862069	1.0	1.0	0.623450	0.597344	0.680032
4	erMax	0.532411	0.862069	1.0	1.0	0.557115	0.504707	0.575797
5	FlipRec	0.577948	1.000000	1.0	1.0	0.736252	0.670221	0.707360

Retained Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.495797	0.918580	0.985714	1.0	0.494855	0.516339	0.541686
1	SISA	0.513901	0.913150	0.978756	1.0	0.582814	0.566368	0.608881
2	RecEraser	0.529526	0.922216	0.985775	1.0	0.524288	0.545700	0.567219
3	BadT	0.512780	0.922756	0.992857	1.0	0.501268	0.525120	0.556548
4	erMax	0.554676	0.926931	0.982143	1.0	0.531807	0.547797	0.579859
5	FlipRec	0.508825	0.914405	0.989286	1.0	0.501611	0.527116	0.555860

Targeted Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.461863	0.800000	1.0	1.0	0.497337	0.415878	0.504283
1	SISA	0.538421	0.966667	0.6	0.0	0.630908	0.425624	0.000000
2	RecEraser	0.503638	0.916667	1.0	1.0	0.482176	0.445528	0.509883
3	BadT	0.581661	0.920000	1.0	1.0	0.546808	0.552935	0.611934
4	erMax	0.492708	0.840000	1.0	1.0	0.492047	0.425571	0.541639
5	FlipRec	0.537532	1.000000	1.0	1.0	0.660795	0.671131	0.739080

Retained Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.508429	0.890244	0.986254	1.0	0.520663	0.535853	0.551031
1	SISA	0.510305	0.921505	1.000000	1.0	0.592296	0.630987	0.772936
2	RecEraser	0.520892	0.895785	0.997059	1.0	0.538169	0.546282	0.563429
3	BadT	0.540428	0.905923	0.986254	1.0	0.552702	0.570900	0.599253
4	erMax	0.559545	0.914634	0.993127	1.0	0.583324	0.589172	0.604902
5	FlipRec	0.506595	0.912892	0.975945	1.0	0.524188	0.536591	0.563213

Targeted Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.534621	0.892857	0.888889	1.000000	0.489268	0.404656	0.443925
1	SISA	0.524206	0.825000	0.933333	1.000000	0.626143	0.605361	0.701778
2	RecEraser	0.473384	0.880000	0.900000	1.000000	0.438076	0.470176	0.600314
3	BadT	0.532840	0.750000	0.888889	0.888889	0.448258	0.450480	0.457720
4	erMax	0.544843	0.892857	0.944444	1.000000	0.464571	0.549326	0.597821
5	FlipRec	0.574560	0.964286	1.000000	1.000000	0.737655	0.751967	0.741113

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NCF Unlearning Effectiveness

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.02	0.07	0.04	0.09	0.11	0.12
1	Targeted (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
2	Retained (Activation)	0.02	0.07	0.04	0.09	0.11	0.11
3	Retained (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
4	All (Activation)	0.02	0.07	0.04	0.09	0.11	0.11
5	All (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.05	0.05	0.16	0.02	0.18	0.20
1	Targeted (JS-Div)	0.00	0.00	0.01	0.00	0.01	0.01
2	Retained (Activation)	0.05	0.05	0.17	0.02	0.18	0.20
3	Retained (JS-Div)	0.00	0.00	0.01	0.00	0.01	0.01
4	All (Activation)	0.05	0.05	0.17	0.02	0.18	0.20
5	All (JS-Div)	0.00	0.00	0.01	0.00	0.01	0.01

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.01	0.01	0.06	0.01	0.10	0.12
1	Targeted (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
2	Retained (Activation)	0.01	0.01	0.07	0.01	0.10	0.12
3	Retained (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
4	All (Activation)	0.01	0.01	0.07	0.01	0.10	0.12
5	All (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00

DMF Results

Retained Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.538328	0.871575	0.979592	0.991525	0.493480	0.514747	0.510650
1	SISA	0.504027	0.920473	0.974411	1.000000	0.574533	0.596374	0.713133
2	RecEraser	0.520396	0.870162	0.979282	1.000000	0.490019	0.501508	0.511623
3	BadT	0.710089	0.929795	0.986395	0.991525	0.662756	0.676986	0.691035
4	erMax	0.714244	0.929795	0.989796	1.000000	0.677902	0.681694	0.700079
5	FlipRec	0.535200	0.875000	0.979592	0.991525	0.500420	0.518293	0.532854

Targeted Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.577722	0.862069	1.000000	1.0	0.591054	0.525027	0.627935
1	SISA	0.537711	0.933333	0.200000	0.0	0.664945	0.159835	0.000000
2	RecEraser	0.509283	0.808333	1.000000	0.4	0.517475	0.552453	0.247455
3	BadT	0.826367	0.896552	1.000000	1.0	0.815950	0.841915	0.948801
4	erMax	0.460849	0.758621	1.000000	1.0	0.497555	0.485911	0.563865
5	FlipRec	0.513771	0.931034	0.909091	1.0	0.666145	0.595908	0.658824

Retained Results Summary:

	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
0	Retrained	0.538328	0.871575	0.979592	0.991525	0.493480	0.514747	0.510650
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	Method	AUC	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
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3	BadT	0.710089	0.929795	0.986395	0.991525	0.662756	0.676986	0.691035
4	erMax	0.714244	0.929795	0.989796	1.000000	0.677902	0.681694	0.700079
5	FlipRec	0.535200	0.875000	0.979592	0.991525	0.500420	0.518293	0.532854

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2	RecEraser	0.509283	0.808333	1.000000	0.4	0.517475	0.552453	0.247455
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5	FlipRec	0.513771	0.931034	0.909091	1.0	0.666145	0.595908	0.658824

DMF Unlearning Effectiveness

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.04	0.04	0.04	0.04	0.04	0.04
1	Targeted (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
2	Retained (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
3	Retained (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
4	All (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
5	All (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.04	0.04	0.04	0.04	0.04	0.04
1	Targeted (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
2	Retained (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
3	Retained (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
4	All (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
5	All (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00

	Group	Retrained	SISA	RecEraser	BadT	erMax	FlipRec
0	Targeted (Activation)	0.04	0.04	0.04	0.04	0.04	0.04
1	Targeted (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
2	Retained (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
3	Retained (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00
4	All (Activation)	0.03	0.04	0.03	0.04	0.03	0.05
5	All (JS-Div)	0.00	0.00	0.00	0.00	0.00	0.00

Unlearning Evaluation

Based on the results of six unlearning methods: Retrained, SISA, RecEraser, BadT, erMax, and FlipRec with two neural recommendation training: NCF and DMF. The model was tested on three datasets, each split into retained and targeted user groups. Performance was measured on these metrics: AUC, HR@5, HR@10, HR@20, NDCG@5, NDCG@10, and NDCG@20.

Detailed Analysis of NCF:

Retained users: Retrained and RecEraser consistently achieved high retention performance. In dataset 1, Retrained HR@10 = 0.986395, RecEraser = 0.982621. In dataset 2, RecEraser HR@10 = 0.987551 and dataset 3 both at HR@10 achieved = 0.979592 and NDCG@20 ≈

0.537. While erMax, also showed strong retention slightly below RecEraser. FlipRec, while slightly behind it mained a solid retention around HR@10 ≈ 0.97-0.98 across datasets, making it a balanced choice.

Targeted Users: FlipRec was most effective at forgetting with NDCG@20 values a lot lower across all datasets: Dataset 1: NDCG@20 = 0.707, Dataset 2: NDCG@20 = 0.713 and Dataset 3: NDCG@20 = 0.747. Other methods like RecEraser and SISA also showed effective forgetting. BadT and erMax were the least effective at forgetting, often with high HR and NDCG values.

Detailed Analysis of DMF:

Retained Users: Retrained, BadT, and erMax achieved extremely good retention. In dataset 1: erMax HR@10 = 0.987952, NDCG@20 = 0.700, dataset 2: BadT NDCG@20 = 0.948, and dataset 3: all methods scored HR@10 = 0.979592 or higher. FlipRec with slightly lower performance but remained consistent.

Targeted Users: FlipRec was most consistent at forgetting with dataset 1: NDCG@20 = 0.658, dataset 2: NDCG@20 = 0.658, and dataset 3: NDCG@20 = 0.705. Retrained, RecEraser, and SISA showed varied results depending on the datasets. BadT and erMax have high NDCG showing it is not as effective at forgetting.

In conclusion, FlipRec is the only method that consistently balances strong retention and is effective at forgetting across both models and datasets. Retrained is best for preserving user information but is very expensive and not suitable for scalable unlearning. BadT and erMax have high retention but are bad at forgetting. SISA and RecEraser show good mid-range performance depending on the datasets.

Unlearning Effectiveness Evaluation

Activation Distance- calculates the average L2 distance between predicted probabilities of unlearned model and retrained model. A higher means the model has changed more after unlearning. JS-Divergence [10]- measure how different the predicted probability distributions are between unlearning and retrained models. Lower values are better at forgetting.

NCF: Unlearning Effectiveness

Targeted Users (Who should be forgotten): FlipRec achieved the highest activation distance in all three datasets (0.12, 0.20, 0.12) meaning it altered its internal state the most when forgetting targeted users. JS-Divergence is extremely low with (0.00-0.01), however FlipRec still shows some non-zero values which show change in output distribution. So, FlipRec is most effective at forgetting users in NCF.

Retained Users (Who should not be affected): FlipRec have activation distance of 0.10-0.12 which is higher than others but still in a good range and JS-Divergence is zero so showing no accidental forgetting of retained users. So, this shows that FlipRec can forget targeted users without harming the retained users.

DMF: Unlearning Effectiveness

Targeted Users: all methods show similar activation distance of 0.04 across all datasets and JS-Divergence is zero for all meaning no change in output distribution. So, DMF all methods perform similarly and unlearning may be less effective overall.

Retained Users: Activation Distance are low and nearly equal for all methods and

JS-Divergence is zero which means there is no accidental forgetting.As such, all methods are safe for retained users.

In conclusion, FlipRec is most effective in the NCF model which makes real internal changes for forgotten users while keeping retained users safe. In DMF, the model is less reactive to forgetting and suggests more tailored unlearning strategies.

IV. Summary

In this paper, we explored the problem of efficient user unlearning in recommendation systems, where removing user data often leads to increased computational cost or reduced accuracy. To address this, we implement FlipRec, a method that flips user interaction labels and fine-tunes the model to forget specific users without retraining from scratch. We applied FlipRec to two models Neural Collaborative Filtering (NCF) and Deep Matrix Factorization (DMF) and compared its performance with five other unlearning methods: Retrained, SISA, RecEraser, BadT and erMax. Using MovieLens-based datasets with binary feedback, we evaluate both retention and forgetting effectiveness through AUC, HR, NDCG, Activation Distance, and JS-Divergence. Results show that FlipRec achieves the best balance in the NCF model; it effectively forgets targeted users while preserving performance for retained ones. In the DMF model, all methods perform similarly, but FlipRec remains competitive. Overall, FlipRec proved to be an efficient and reliable approach for unlearning in recommendation systems.

V. Teamwork

Tam Hoa Vo wrote all the code for unlearning models, code for HR and NDCG, splitting the data and wrote the paper. Adila Biswas wrote the code for DMF and the unlearning effectiveness code, helped edit the paper, and presentation slides. Bahaar Grewal wrote the code for NCF and AUC, ran all the datasets on the code/adding comments, helped edit the paper, and presentation slides.

VI. Reference

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